

Subject: Urgent Compliance Violation and Directive to Halt All External Data Transfers

To: Dr. Chen, Lead Data Scientist; Dr. Sharma, Clinical Data Coordinator; Mr. Davies, IT Security & Compliance Officer; All Project Personnel
Cc: University Legal Counsel; Research Integrity Office

This memorandum constitutes a formal notification of a critical and unacceptable breach of the Multimodal AI-AD project's foundational data governance protocols. It has come to my attention, via the quarterly infrastructure audit, that high-resolution 3D T1-weighted MRI scans sourced from the Knight Alzheimer's Disease Research Center (ADRC) cohort are currently being processed for volumetric segmentation on the computational servers of "NeuroCloud Analytic Solutions," a third-party commercial provider.

This activity is being conducted in the absence of a formally executed Data Processing Agreement (DPA) as required under Article 28 of the General Data Protection Regulation (GDPR). This is a severe contractual and regulatory violation that exposes this institution, our funding bodies, and most gravely, our research participants, to significant legal and ethical risk.

The justification provided by the engineering team—that the data is "anonymous"—demonstrates a profound and dangerous misunderstanding of both technical reality and legal definitions, which I will now clarify authoritatively.

1. The Critical Distinction: Pseudonymization vs. Anonymization

- Pseudonymization (Our Protocol): This is a reversible data management process where direct identifiers (name, NHS/SSN, etc.) are replaced with a project-specific key code. The identifying data is kept separately under strict controls. The data itself remains personal data under GDPR. Our neuroimaging data is pseudonymized. Each scan is linked to a participant ID, which in turn is linkable to a master key held by our sanctioned Data Custodian.
- Anonymization (A Legal Standard): This is an irreversible process by which personal data is stripped of all identifiable elements to such a degree that the data subject can never be re-identified by any means reasonably likely to be used. For high-dimensional data like 3D brain scans, true anonymization is virtually impossible.

2. The Re-identification Risk in Neuroimaging

A 3D structural MRI is not an anonymous blob of data. It is a unique, high-fidelity biometric map. Research has consistently shown that the cortical folding patterns, volumetric structures, and even functional connectivity fingerprints are sufficiently

distinct to allow re-identification through linkage attacks, especially when combined with other datasets. To claim these scans are "anonymous" is scientifically negligent. They are highly sensitive, pseudonymized personal data concerning health, afforded the highest level of protection under GDPR (Article 9).

3. The Legal Imperative of a Data Processing Agreement

Without a DPA, "NeuroCloud Analytic Solutions" is an uncontrolled data processor. We have no contractual guarantee of:

- Their sub-processor governance.
- Their technical and organizational security measures (TOMs).
- Their data retention and deletion schedules.
- Their protocols for reporting a data breach.
- Their adherence to GDPR's principles of data minimization and purpose limitation.

Their processing of our data is, in the eyes of the law, unauthorized. This jeopardizes not only our ethical standing but the entire legal basis for processing (Lawful Basis: Public Task; Special Category Condition: Article 9(2)(j) – scientific research).

DIRECTIVE & IMMEDIATE ACTIONS:

1. CEASE & DESIST: Effective 1700 GMT today, all transfer of project data to any external computational resource, including "NeuroCloud Analytic Solutions," is hereby suspended. Any automated pipelines must be disabled.
2. DATA RECALL: Mr. Davies, in coordination with Dr. Chen, you will oversee the secure, verifiable deletion of all Knight ADRC-derived MRI data from the third-party servers. You will obtain and archive a certificate of deletion from the provider. All local derived data from this unauthorized processing is to be quarantined.
3. PIA FINALIZATION: Processing will not resume until the University's Data Protection Officer has finalized the comprehensive Privacy Impact Assessment (PIA) for this specific use-case and provider. The PIA must detail the re-identification risks and the mitigating controls encoded in the DPA.
4. CONTRACTUAL REMEDIATION: Legal Counsel will immediately commence negotiation of a GDPR-compliant DPA with "NeuroCloud Analytic Solutions," incorporating the stringent controls mandated by our PIA. No data will flow until I have personally approved the executed agreement.

Scientific Integrity and Publication Consequence:

This lapse is not merely bureaucratic. "Nature Medicine," our target journal, requires explicit statements on ethical approval, data provenance, and GDPR/compliance in the Methods section. Any ambiguity here would be grounds for immediate editorial

rejection and could trigger an institutional investigation, irreparably damaging the project's credibility and our ability to recruit future cohorts. Our science must be unimpeachable from the data layer upward.

This is non-negotiable. Compliance with data protection law is not an obstacle to research; it is the framework that permits ethical, reputable, and ultimately publishable research to occur. Your immediate cooperation is required.

Dr. Alice Robertson, Ph.D., F.R.S.M.
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Chair, Neurocomputational Ethics Advisory Board